

# Anuj Chauhan

Ph.D. National Taiwan University (國立臺灣大學) · Taipei City, Taiwan

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Taiwan Employment Gold Card Holder (台灣就業金卡)

## PROFESSIONAL SUMMARY

**Industrial Simulation Engineer** delivering production-validated **physics-based simulation** solutions at Pegatron Corp. I design and deploy **CFD (fluid dynamics)** and **FEA (stress, deformation)** models across the full ANSYS suite — **Fluent, Mechanical, Icepak, LS-DYNA, and Twin Builder** — to solve real thermal, structural, multiphase, and **data center cooling** challenges in high-reliability hardware. Specialized in building **Digital Twins** and **Reduced Order Models (ROMs)** that turn complex 3D simulations into rapid, decision-ready insights. Backed by a **Ph.D. from National Taiwan University**, hands-on semiconductor nanofabrication expertise (EBL, photolithography, RIE/ICP, full cleanroom integration). Bridging the rare gap between **nanoscale fabrication** and **industrial-scale simulation** — ready to apply both to advanced semiconductor manufacturing.

## CORE COMPETENCIES & TECH STACK

| Semiconductor Fabrication & Metrology                                                                                                                                                                                                                                                                                                                                                                                      | Process Simulation & Modeling                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <ul style="list-style-type: none"><li><b>Lithography:</b> Electron Beam Lithography (EBL), Advanced Photolithography, PR Optimization, CD Control.</li><li><b>Cleanroom Processing:</b> End-to-end Process Integration, PVD/CVD, ALD, Plasma Etching (RIE/ICP), Wet Processing.</li><li><b>Metrology:</b> SEM, TEM, Optical Inspection, Defect Analysis, Yield Optimization, Critical Dimension (CD) Validation.</li></ul> | <ul style="list-style-type: none"><li><b>Focus:</b> Multi-Physics Modeling (Thermal/Stress), Data Center Cooling &amp; Thermal Management, Structural Reliability, Physics-Based Digital Twins, DOE, ROM.</li><li><b>Stack:</b> ANSYS (Mechanical, Fluent Multiphase VOF, Icepak, Heatsink, LS-DYNA, Twin Builder, EnSight, ParaView), PyTwin, NVIDIA Omniverse, PyANSYS, SolidWorks, SpaceClaim.</li><li><b>Industry Proven:</b> All simulation capabilities applied and validated on real industrial projects at Pegatron Corp — from concept to production-ready digital twin deployment.</li></ul> |

## PROFESSIONAL EXPERIENCE

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| <b>PEGATRON CORPORATION</b><br><i>Section Manager, Software R&amp;D</i><br><i>Technical Manager, Software R&amp;D</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Taipei, Taiwan</i><br><b>Mar 2025 – Present</b><br><b>Jun 2024 – Mar 2025</b> |
| <ul style="list-style-type: none"><li><b>Multi-Physics Simulation &amp; Reliability:</b> Executed advanced ANSYS Mechanical and Fluent simulations for industrial projects, performing high-fidelity stress, deformation, and CFD analysis to optimize structural integrity and fluid flow.</li><li><b>LS-DYNA Simulation:</b> Developed non-linear FEA models in LS-DYNA for large-scale deformation and crash analysis, integrating simulation outputs directly into digital twin workflows.</li><li><b>Data Center Thermal &amp; Cooling Simulation:</b> Developed high-fidelity CFD models using ANSYS Fluent and Icepak to simulate airflow, heat dissipation, and cooling efficiency across data center rack and server environments; delivered actionable insights for thermal management optimization and energy efficiency improvements.</li><li><b>Reduced Order Modeling (ROM):</b> Engineered computationally efficient models using ANSYS Twin Builder to compress complex 3D FEA/CFD results for rapid, accurate reliability assessment.</li><li><b>Digital Twin Pipeline:</b> Spearheaded development of physics-based simulation workflows enabling predictive monitoring of industrial hardware performance under varying structural and thermal constraints.</li></ul> |                                                                                  |
| <b>NATIONAL TAIWAN UNIVERSITY</b><br><i>(Grad. Institute of Photonics &amp; Optoelectronics) — Postdoctoral Fellow</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <i>Taipei, Taiwan</i><br><b>Apr 2023 – May 2024</b>                              |

- **Advanced Lithography & Nanofabrication:** Designed and patterned ultra-compact plasmonic waveguides and optical ring-resonators, achieving strict nanoscale tolerances using Electron Beam Lithography (EBL) and precision Photolithography.
- **Process Integration & Optimization:** Directed comprehensive cleanroom process flows (Deposition, Patterning, Etching). Optimized photoresist profiles and plasma etching recipes (RIE/ICP) to significantly improve device yield and reliability.
- **Metrology & Characterization:** Conducted rigorous inspection using SEM and optical metrology tools to validate Critical Dimensions (CD) and ensure high process fidelity across integration cycles.

#### SHARDA UNIVERSITY

(Department of Physics & Environmental Sciences) — Research Assistant

Greater Noida, India

Jan 2018 – Dec 2018

- Fabrication and characterization of dye-sensitized solar cells and supercapacitors; developed hands-on materials processing and electrochemical characterization skills.

#### EDUCATION

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##### National Taiwan University

Ph.D., Chemistry

Award: NTU Outstanding International Graduate Student Scholarship

Taipei, Taiwan

2019 – 2023

##### Gautam Buddha University

M.Sc., Applied Physics

Greater Noida, India

2014 – 2016

##### University of Delhi

B.Sc., Physics

New Delhi, India

2010 – 2013